CS3 RESEARCH EXPERIENCE FOR TEACHERS 2024

Perspectives in Philosophy of Science

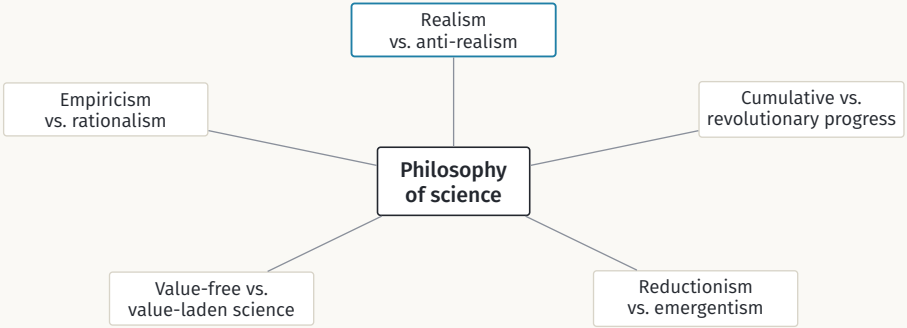
Realism, anti-realism, and beyond

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Five divisions that organize the field



We will explore a range of perspectives through these lenses, starting with the first: do our theories describe reality?

The core debate: realism versus anti-realism

Key question

Do our best scientific theories truly describe an independent reality?

Realist perspectives

Scientific realism
Ontic structural realism

Anti-realist perspectives

Instrumentalism
Constructive empiricism

Mixed or nuanced perspectives

Critical rationalism
Relativism
Pragmatism

The next seven slides take these positions one at a time, in this order.

Scientific realism: theories describe the world as it is

Key idea

Science aims to give true descriptions of both the observable and the unobservable aspects of the world.

Core principles

1. Scientific theories are approximately true descriptions of reality.
2. The terms of mature scientific theories refer to real entities.
3. Science makes progress by getting closer to the truth.

Main proponents

Hilary Putnam (1926–2016)
Richard Boyd (1942–2021)

View on scientific progress

An increasingly close approximation to the truth about observable and unobservable reality.

Popularity

Became the dominant view in the second half of the 20th century.

Ontic structural realism: theories reveal structure

Key idea

Scientific theories reveal the fundamental structure of a mind-independent reality.

Core principles

1. Scientific theories describe a mind-independent reality.
2. The ontology of fundamental physics reflects the true nature of reality.
3. The emphasis falls on the structural aspects of theories.

Main proponents

James Ladyman (1967–present)
Steven French (1959–present)

View on scientific progress

Uncovering more of reality's fundamental structure.

Popularity

Gained traction in the late 20th and early 21st centuries, especially in the philosophy of physics.

Instrumentalism: theories are tools for prediction

Key idea

Scientific theories are tools for prediction, not descriptions of reality.

Core principles

1. Theories are instruments for making predictions, not claims to truth.
2. Agnostic about unobservables.
3. The focus is on empirical adequacy and practical usefulness.

Main proponents

John Dewey (1859–1952)
Pierre Duhem (1861–1916)

View on scientific progress

Developing more useful predictive tools.

Popularity

Gained traction in the early 20th century and remains influential in some circles.

Constructive empiricism: aim for empirical adequacy

Key idea

Science aims for empirical adequacy, not for truth about unobservables.

Core principles

1. The aim of science is empirical adequacy, not truth.
2. Agnostic about unobservables.
3. We can accept and use theories without believing them to be true.

Main proponents

Bas van Fraassen (1941–present)

View on scientific progress

Improving the empirical adequacy of theories.

Popularity

Has drawn attention since the 1980s as a middle ground between realism and instrumentalism.

Critical rationalism: conjecture and refutation

Key idea

Scientific progress happens through conjecture and refutation.

Core principles

1. Falsificationism: theories should make risky predictions that can be tested.
2. We can never verify a theory, only falsify it.
3. Scientific knowledge is always tentative.

Main proponents

Karl Popper (1902–1994)

View on scientific progress

Conjecture and refutation, eliminating false theories along the way.

Popularity

Rose to prominence in the mid-20th century and has been widely influential in the philosophy of science.

Relativism: knowledge depends on its context

Key idea

Scientific knowledge and truth are relative to historical, cultural, or theoretical contexts.

Core principles

1. Scientific paradigms determine what counts as evidence and as truth.
2. Paradigms are incommensurable with one another.
3. Social and historical factors shape scientific knowledge.

Main proponents

Thomas Kuhn (1922–1996)
Paul Feyerabend (1924–1994)

View on scientific progress

Paradigm shifts, not cumulative growth.

Popularity

Gained significant influence in the 1960s and 1970s and remains controversial.

Pragmatism: judge theories by what they do

Key idea

Scientific theories should be judged by their practical consequences and their ability to solve problems.

Core principles

1. Truth is what works in practice.
2. Scientific inquiry should focus on solving real-world problems.
3. Theories are tools for prediction and control.

Main proponents

Charles Sanders Peirce (1839–1914)
William James (1842–1910)
John Dewey (1859–1952)

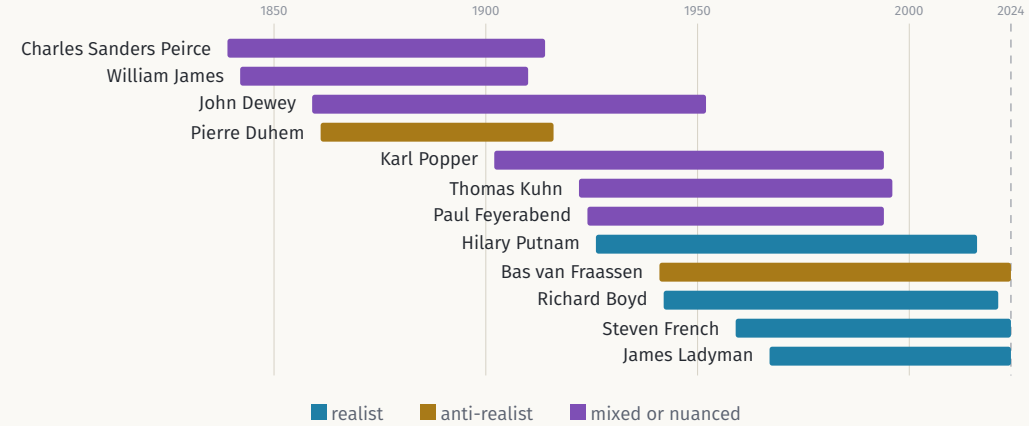
View on scientific progress

Solving more complex problems and improving the human condition.

Popularity

Influential in the late 19th and early 20th centuries, and experiencing a resurgence in contemporary philosophy of science.

The proponents on one timeline



Bars for living philosophers end at 2024. Dewey, named for both pragmatism and instrumentalism, is colored for pragmatism.

Seven positions at a glance

Position	Aim or key idea	View on progress
Scientific realism	True descriptions of observables and unobservables	Closer approximation to truth
Ontic structural realism	The fundamental structure of reality	Uncovering more of that structure
Instrumentalism	Theories as tools for prediction	More useful predictive tools
Constructive empiricism	Empirical adequacy, not truth	Improved empirical adequacy
Critical rationalism	Conjecture and refutation	Eliminating false theories
Relativism	Truth relative to context	Paradigm shifts
Pragmatism	Judged by practical consequences	Solving more complex problems

Empiricism versus rationalism

Empiricist leanings

Positivism: all knowledge comes from sensory experience.

Constructive empiricism: focus on observable phenomena.

Instrumentalism: theories are judged by their empirical predictions.

Rationalist elements

Critical rationalism: emphasis on the logical structure of theories.

Scientific realism: rational inference to unobservables.

Ontic structural realism: rational access to fundamental reality.

Mixed approaches

Pragmatism: combines empirical testing with practical reasoning.

Relativism: emphasizes how theoretical frameworks shape empirical observations.

Views on scientific progress

Cumulative progress

Positivism: accumulation of verified knowledge.

Scientific realism: closer approximation to the truth.

Ontic structural realism: uncovering more of reality's structure.

Non-cumulative or revolutionary

Relativism: paradigm shifts (Kuhn).

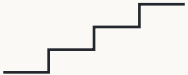
Critical rationalism: eliminating false theories.

Progress as improvement

Instrumentalism: more useful predictive tools.

Constructive empiricism: improving empirical adequacy.

Pragmatism: solving more complex problems.



adding to what is known



a shift between frameworks



tools that work better

Reductionism versus emergentism

Reductionist tendencies

Positivism: the unity of science, with everything reducible to physics.

Some forms of scientific realism: a fundamental physical ontology.

Emergentist or anti-reductionist

Some ontic structural realists: emergent properties at different scales.

Pragmatism: different levels of explanation for different purposes.

Some relativists: different paradigms or frameworks are irreducible.

Mixed or nuanced positions

Contemporary scientific realism: acknowledges emergent properties while keeping reducibility in principle.

Critical rationalism: focuses on problem-solving rather than reduction.

KEY DEBATES

Seven open questions in philosophy of science

1. **Realism versus anti-realism**: do our best scientific theories truly describe an independent reality?
2. **The demarcation problem**: what distinguishes science from non-science?
3. **Theory choice**: how do scientists choose between competing theories?
4. **The problem of induction**: can we justify inductive reasoning in science?
5. **Values in science**: what role should values play in scientific inquiry?
6. **Reductionism versus emergentism**: can all sciences be reduced to physics?
7. **Scientific explanation**: what makes a good scientific explanation?